

Vim & Nano Editor Commands Guide

A practical reference to Vim and Nano text editors: modes, commands, and how they work

Part 1: Vim Editor

Vim (Vi IMproved) is a mode-based text editor available on almost every Linux/Unix system. Unlike simple editors, Vim works through different **modes** — you switch modes to type text, move around, or run commands. Understanding modes is the key to understanding how Vim commands work.

How Vim Modes Work

Vim opens in **Normal mode** by default. In this mode, keys are not typed as text — they act as commands (move cursor, delete, copy, etc). You switch to **Insert mode** to type text, and press **Esc** to return to Normal mode. **Command mode** (also called Command-line mode) is entered by typing **:** from Normal mode, and is used for saving, quitting, search-and-replace, and other line commands. **Visual mode** lets you select text before acting on it.

Opening Vim

Command	What it does
<code>vim file.txt</code>	Open file.txt in Vim (creates it if it doesn't exist)
<code>vim +12 file.txt</code>	Open file.txt with cursor on line 12
<code>vim -R file.txt</code>	Open file.txt in read-only mode
<code>vimdiff f1 f2</code>	Open two files side by side and highlight differences

Switching Modes

Key	Mode switch / effect
Esc	Return to Normal mode from any other mode
i	Insert mode, before the cursor
a	Insert mode, after the cursor
I	Insert mode, at the start of the line
A	Insert mode, at the end of the line
o	Insert mode, on a new line below current
O	Insert mode, on a new line above current
v	Visual mode (character-wise selection)
V	Visual Line mode (whole-line selection)
Ctrl + v	Visual Block mode (column selection)
:	Command-line mode (for :w, :q, etc.)

Moving the Cursor (Normal mode)

Command	What it does
h / j / k / l	Move left / down / up / right, one character/line at a time
w	Jump to the start of the next word
b	Jump to the start of the previous word

Command	What it does
e	Jump to the end of the current/next word
0	Jump to the start of the line
\$	Jump to the end of the line
gg	Go to the first line of the file
G	Go to the last line of the file
12G or :12	Go to line 12
Ctrl + f	Scroll forward one full page
Ctrl + b	Scroll backward one full page
%	Jump to the matching bracket () { } []

Editing Text (Normal mode)

Command	What it does
x	Delete the character under the cursor
dd	Delete (cut) the current line
3dd	Delete 3 lines starting from the current one
dw	Delete from cursor to the start of the next word
D	Delete from cursor to the end of the line
yy	Yank (copy) the current line
3yy	Yank 3 lines
p	Paste after the cursor / below the current line
P	Paste before the cursor / above the current line
u	Undo the last change
Ctrl + r	Redo the last undone change
r<char>	Replace one character under cursor with <char>
cw	Change (delete + insert) to end of word
.	Repeat the last change

Search & Replace

Command	What it does
/pattern	Search forward for 'pattern'
?pattern	Search backward for 'pattern'
n	Repeat the search in the same direction
N	Repeat the search in the opposite direction
:%s/old/new/g	Replace every 'old' with 'new' in the whole file
:%s/old/new/gc	Same, but ask for confirmation on each match

Saving & Exiting

Command	What it does
<code>:w</code>	Save (write) the file
<code>:w newname.txt</code>	Save a copy as newname.txt
<code>:q</code>	Quit (fails if there are unsaved changes)
<code>:q!</code>	Quit without saving, discarding changes
<code>:wq</code> or <code>:x</code>	Save and quit
<code>ZZ</code>	Save and quit (Normal mode shortcut)
<code>:wqa</code>	Save and quit all open files/splits

Tip: Vim commands generally follow the pattern **[count] command [motion]**. For example, **3dw** means "delete 3 words", and **2dd** means "delete 2 lines". Learning motions (w, b, e, \$, 0, gg, G) lets you combine them with any operator (d = delete, y = yank, c = change) to build new commands.

Part 2: Nano Editor

Nano is a simple, beginner-friendly command-line text editor. Unlike Vim, Nano has no separate modes — you type text directly, and commands are run using **Ctrl** and **Alt (Meta)** key combinations, which are always shown at the bottom of the screen as a cheat sheet (the caret ^ symbol means Ctrl).

How Nano Commands Work

Every Nano command is a key combination: hold **Ctrl** and press a letter (shown as ^ followed by the letter, e.g. ^O means Ctrl+O), or press **Alt/Esc** then a letter (shown as M-, e.g. M-U means Alt+U). Because there is no mode switching, whatever you type appears in the document immediately — commands only fire on the special key combinations.

Opening Nano

Command	What it does
nano file.txt	Open file.txt in Nano (creates it if it doesn't exist)
nano +12 file.txt	Open file.txt with cursor on line 12
nano -v file.txt	Open file.txt in read-only (view) mode
nano -l file.txt	Open with line numbers shown

File Commands

Command	What it does
Ctrl + O	Write Out (save) the file
Ctrl + X	Exit Nano (prompts to save if there are changes)
Ctrl + R	Read/insert another file into the current one
Ctrl + S	Save the file without exiting (in newer Nano versions)

Moving the Cursor

Command	What it does
Arrow keys	Move the cursor up/down/left/right
Ctrl + A	Move to the beginning of the line
Ctrl + E	Move to the end of the line
Ctrl + Y	Move up one page (Page Up)
Ctrl + V	Move down one page (Page Down)
Ctrl + _	Go to a specific line and column number
Alt + \	Go to the first line of the file
Alt + /	Go to the last line of the file

Editing Text

Command	What it does
Ctrl + K	Cut the current line (cut to buffer)
Ctrl + U	Paste (uncut) the last cut text
Alt + 6	Copy the current line without cutting it
Ctrl + Shift + 6 (or Alt+A)	Start text selection (mark)
Ctrl + 6	Set a selection mark, then move to select text
Ctrl + D	Delete the character under the cursor
Backspace	Delete the character before the cursor
Ctrl + Z	Undo the last action
Ctrl + Shift + Z (or Alt+U)	Redo the last undone action
Tab	Insert a tab / indent selected text

Search, Replace & Help

Command	What it does
Ctrl + W	Search (Where Is) for text
Alt + W	Repeat the last search (find next)
Ctrl + \	Search and replace text
Ctrl + G	Show the built-in help screen

Tip: Nano always shows the most useful shortcuts on the last two lines of the terminal window, so you rarely need to memorize the full list — glance at the bottom bar whenever you forget a command.

Last Two Tasks: Practice Walkthroughs

Task 1 - Vim: Edit and Save a File

Step	Command	Result
1	vim notes.txt	Opens notes.txt in Normal mode
2	i	Switches to Insert mode
3	Type your text	Text is typed into the file
4	Esc	Returns to Normal mode
5	:wq	Saves the file and exits Vim

Task 2 - Nano: Edit and Save a File

Step	Command	Result
1	nano notes.txt	Opens notes.txt, ready to type immediately
2	Type your text	Text appears directly, no mode switch needed
3	Ctrl + O	Prompts to confirm the filename, then writes the file
4	Enter	Confirms the filename and saves
5	Ctrl + X	Exits Nano

Quick comparison: Vim requires switching between Normal and Insert modes (i to type, Esc to stop, :wq to save+quit), which feels harder at first but is very fast once memorized. Nano has no modes — you type immediately and use Ctrl-key shortcuts shown on-screen (Ctrl+O to save, Ctrl+X to exit), making it easier for beginners.